

Aum Amriteswaryai Namaha



Major Project 21CSA399A

Interim Report

Project Title: Fake News Detection using Deep Learning

Student Name: Harikrishnan R

Roll No.: AA.SCU3CSC2107088

Abstract

The rapid growth of online news platforms and social media has increased the spread of misinformation and fake news. Fake news can influence public opinion, create social unrest, and mislead individuals by presenting false or manipulated information as genuine news. Manual verification of news articles is time-consuming and difficult due to the large volume of content generated every day.

This project aims to develop a Fake News Detection System using Deep Learning and Natural Language Processing (NLP) techniques. The proposed system will analyze the textual content of news articles and classify them as fake or genuine. The system will utilize the Fake News Dataset obtained from Kaggle and employ a Long Short-Term Memory (LSTM) neural network model for classification.

The project involves text preprocessing, tokenization, feature extraction, word embedding, model training, and evaluation. Performance will be measured using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The proposed solution aims to demonstrate how artificial intelligence can assist in identifying misinformation and support reliable information dissemination in digital platforms.

1. Objectives

- 1.1. To study the application of Deep Learning techniques in fake news detection.
- 1.2. To preprocess and analyze textual news data using Natural Language Processing techniques.
- 1.3. To design and implement an LSTM-based classification model for detecting fake news.
- 1.4. To evaluate the performance of the model using standard classification metrics.
- 1.5. To develop a user-friendly interface for news classification.

2. Scope

The scope of this project is limited to textual fake news detection using supervised deep learning techniques. The project focuses on analyzing the content of news articles and classifying them as fake or genuine based on linguistic patterns and contextual information.

The study covers dataset collection, preprocessing, tokenization, feature extraction, model development, training, testing, and evaluation. The proposed system will use publicly available datasets and will operate on English-language news articles.

The project does not focus on image-based misinformation, video analysis, social network propagation analysis, or real-time fact-checking using external sources. Instead, it concentrates on text-based classification using Natural Language Processing and Deep Learning.

The outcome of the project will be a trained classification model capable of predicting the authenticity of news articles and demonstrating the practical application of AI in combating misinformation.

3. Introduction

The internet has become one of the primary sources of information for people around the world. While this has improved accessibility to information, it has also enabled the rapid spread of fake news. Fake news refers to intentionally misleading or false information presented as legitimate news content.

The rise of social media platforms has accelerated the dissemination of misinformation, making manual verification increasingly difficult. Automated fake news detection systems have therefore become an important area of research in Artificial Intelligence and Natural Language Processing.

Machine Learning and Deep Learning techniques have shown promising results in text classification tasks. In recent years, neural network architectures such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have been successfully applied to sentiment analysis, spam detection, and fake news classification.

This project aims to leverage Deep Learning techniques to automatically identify fake news articles by analyzing textual content. The proposed system will process news articles, extract meaningful features, and classify them into fake or genuine categories using an LSTM-based model.

3.1. Background

Fake news has emerged as a major challenge in the digital age. The increasing dependence on online news sources has made it easier for misleading information to reach a large audience within a short period of time.

Traditional methods of news verification rely heavily on human experts and fact-checking organizations. However, the enormous volume of content generated daily makes manual verification inefficient and time-consuming.

Recent advances in Artificial Intelligence and Natural Language Processing have enabled automated systems to identify linguistic patterns commonly associated with fake news. Deep Learning models can learn contextual relationships within text and achieve high accuracy in classification tasks.

The proposed project seeks to utilize these advancements to create a practical fake news detection system capable of assisting users in identifying potentially misleading content.

3.2. Related Works/Existing System

Several studies have explored the use of Machine Learning and Deep Learning techniques for fake news detection.

Traditional approaches commonly use algorithms such as Logistic Regression, Naïve Bayes, and Support Vector Machines with TF-IDF feature extraction. These methods have demonstrated reasonable performance but often struggle to capture contextual relationships between words.

More recent research has focused on Deep Learning architectures such as CNNs, RNNs, and LSTM networks. LSTM models are particularly effective because they can retain contextual information across sequences of words and capture long-term dependencies in textual data.

Although significant progress has been made, challenges remain due to evolving language patterns, dataset quality, and domain-specific variations. This project aims to address these challenges by implementing an LSTM-based classification model and evaluating its effectiveness on publicly available datasets.

3.3. Detailed Problem Statement

The widespread dissemination of fake news through digital platforms poses a serious challenge to society. Manual verification methods are insufficient to handle the growing volume of online content.

There is a need for an automated system capable of analyzing news articles and determining whether they are genuine or fake. The system should be able to process textual content efficiently and provide reliable classification results.

This project seeks to develop a Deep Learning-based fake news detection system that can classify news articles using Natural Language Processing and LSTM neural networks.

3.4. Methods/Algorithms

The proposed methodology consists of the following stages:

3.4.1. Dataset Collection

- Fake News Dataset from Kaggle.

3.4.2. Data Preprocessing

- Lowercasing
- Removal of punctuation
- Stop-word removal
- Tokenization

3.4.3. Feature Extraction

- Word Embedding
- Sequence generation

3.4.4. Model Development

- Long Short-Term Memory (LSTM) Neural Network

3.4.5. Model Evaluation

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

3.4.6. User Interface Development

- Simple interface for entering news content and viewing predictions.

3.5. Expected Results

The expected outcome of the project is a Deep Learning-based Fake News Detection System capable of classifying news articles as fake or genuine with satisfactory accuracy.

The system is expected to demonstrate the effectiveness of Natural Language Processing and LSTM networks for text classification tasks. The final implementation will provide prediction results along with evaluation metrics to measure model performance.

The project is also expected to improve understanding of Deep Learning, Natural Language Processing, and practical AI system development.

Date	14-06-2026
Student Name and Signature	Harikrishnan R